

IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: To verify Stefan's law by electric method.		
EXPERIMENT NO. 04	DEPT. : Physics	ISSUE DATE: 12 August, 2026
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AIM: To verify Stefan's law by electric method APPARATUS: 6V battery, D.C. Voltmeter (0-10V), D.C. Ammeter (0-1A), Electric Bulb (tungsten filament), Rheostat(100ohm). Usually, all the above components are arranged in a single cabinet. THEORY AND FORMULA USED: For black bodies we know that the Stefan's law is $E = \sigma(T^4 - T_0^4)$ where E is the net amount of radiation emitted per second per unit area by a body at temperature T and surrounded by another body at temperature T₀, σ is called Stefan's constant. A similar relation can also hold for bodies that are not black bodies. In such cases we can write $P = C(T^\alpha - T_0^\alpha)$ where P is the total power emitted by a body at temperature T surrounded by another body at temperature T₀, α is a power quite close to 4 and C is some constant depending on the material and area of such a body. Further the relation can be put as, $P = CT^\alpha \left(1 - \frac{T_0^\alpha}{T^\alpha} \right)$ If T >> T₀ (e.g. T=1500K and T₀=300K), we can write $P = C(T^\alpha)$ $\log_{10}P = \alpha \log_{10}T + \log_{10}C$ The graph between $\log_{10}P$ and $\log_{10}T$ should be a straight line whose slope gives. Therefore, to verify		

Stefan's law, we must measure the following two parameters:

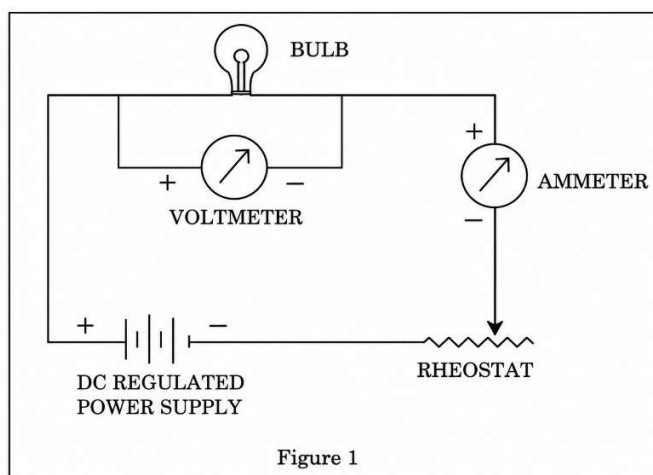
- (i) Power radiated P. We use in this experiment tungsten bulb as the radiating body and in the steady state the electrical power V/I should be equal to the radiating power P (Neglecting power lost in the leads and through the gas in the bulb).
- (ii) Temperature of the radiating body, T. In our experiment we need to measure the temperature of tungsten filament. This is achieved by measuring the resistance of filament, R_t , and then using the relation,

$$R_t = R_0(1 + \alpha t + \beta t^2)$$

With α and β being known for tungsten, we can find

temperature of the filament. One serious limitation arises in the measurement of R_0 (resistance of filament at 0 C or 273 K, one can write R_0 or R_{273K}). At very small current, V/I ratio will give filament resistance as well as lead resistance. To calculate R_0 then we measure the resistance R_g at the stage when the filament first starts glowing and temperature at this stage is approximately 800K. At this temperature contribution of lead resistance becomes smaller by a factor of 4 as computed with that at 273K.

Therefore, measurement of $R_g (=R_{800})$ and using the computed factor R_{800}/R_{273} . A more accurate value of $R_{273}(R_0)$ is obtained.



PROCEDURE:

1. Make the connection as shown in the figure. To connect voltmeter across the bulb, the two wires are soldered to the base points of the bulb.
2. With different increasing and decreasing values of current, we adjust such that the bulb glows each time. Then for every value of V and I, ratio V/I is found which gives R_g . This is the filament resistance at 800k. From R_T/R_0 vs t graph we note that

$$\frac{R_{800}}{R_{273}} = 3.9$$

Therefore, $\frac{R_{800}}{3.9} = R_{273}$

OR, $R_0 = \frac{R_g}{3.9}$

3. Now filament current I is increased from a value below glow stage to values high enough to get dazzling white light, measuring voltage V across bulb every time. From these V and I value, we deduce power $P(=VI)$ and $R_t(=V/I)$. From R_t using the value $R_0(=R_g/3.9)$ or R_{273} , we deduce the temperature t of the filament and obtains graph in $\log_{10}P$ against $\log_{10}T$.

OBSERVATIONS:

1. Least count of Voltmeter:
2. Least count of Ammeter:
3. Glowing resistance R_g :
4. R_0 :

OBSERVATION TABLE:

Determination of Power dissipated P for different temperatures T .

S. NO.	Potential Difference (V)	Current I (A)	Resistance $R_t=V/I$ (Ω)	R_t/R_0	Temperature $T = b \left[\frac{R_t}{R_0} \right] + a$ (K)	$\log_{10}T$	Power $P=VI$ (W)	$\log_{10}P$
1.								
2.								
3.								
4.								
5.								
6.								
7.								
8.								
9.								
10.								

RESULT:

The graph of $\log_{10}P$ vs $\log_{10}T$ comes out to be a straight line. Hence ($P=CT^4$) law is verified. Further the slope of the line as obtained from the graph is ____ and therefore the law is verified as a fourth power law.

PRECAUTIONS:

1. All the measurements should be taken by observing apparatus and readings vertically.
2. Before starting the experiment, all the connections should be made properly.
3. The apparatus should be switched off after taking readings.
4. The glowing resistance should be taken very efficiently.
5. Resistance at a particular temperature should be taken for every setting of voltage and current but the value of R_0 should be fixed for the whole experiment.

Viva Questions

Q1. What is Stefan's Law?

Q2. What is a Black Body?

Q3. What happens to the resistance if the temperature is raised?

Q4. What is R_g and R_0 ?

Q5. What would be the form of Stefan's law for a non-perfect black body?

Table for variation of R/R_0 vs Temperature (T) for Tungsten filament lamp

Temperature (T) °K	R_t/R_0	Temperature (T) °K	R_t/R_0
273	1.00	1373	7.60
373	1.53	1473	8.26
473	2.07	1573	8.90
573	2.13	1673	9.70
673	3.22	1773	10.43
773	3.80	1873	11.17
873	4.40	1973	11.42
973	5.00	2073	12.67
1073	5.64	2173	13.50
1173	6.37	2273	14.30
1273	6.94		

